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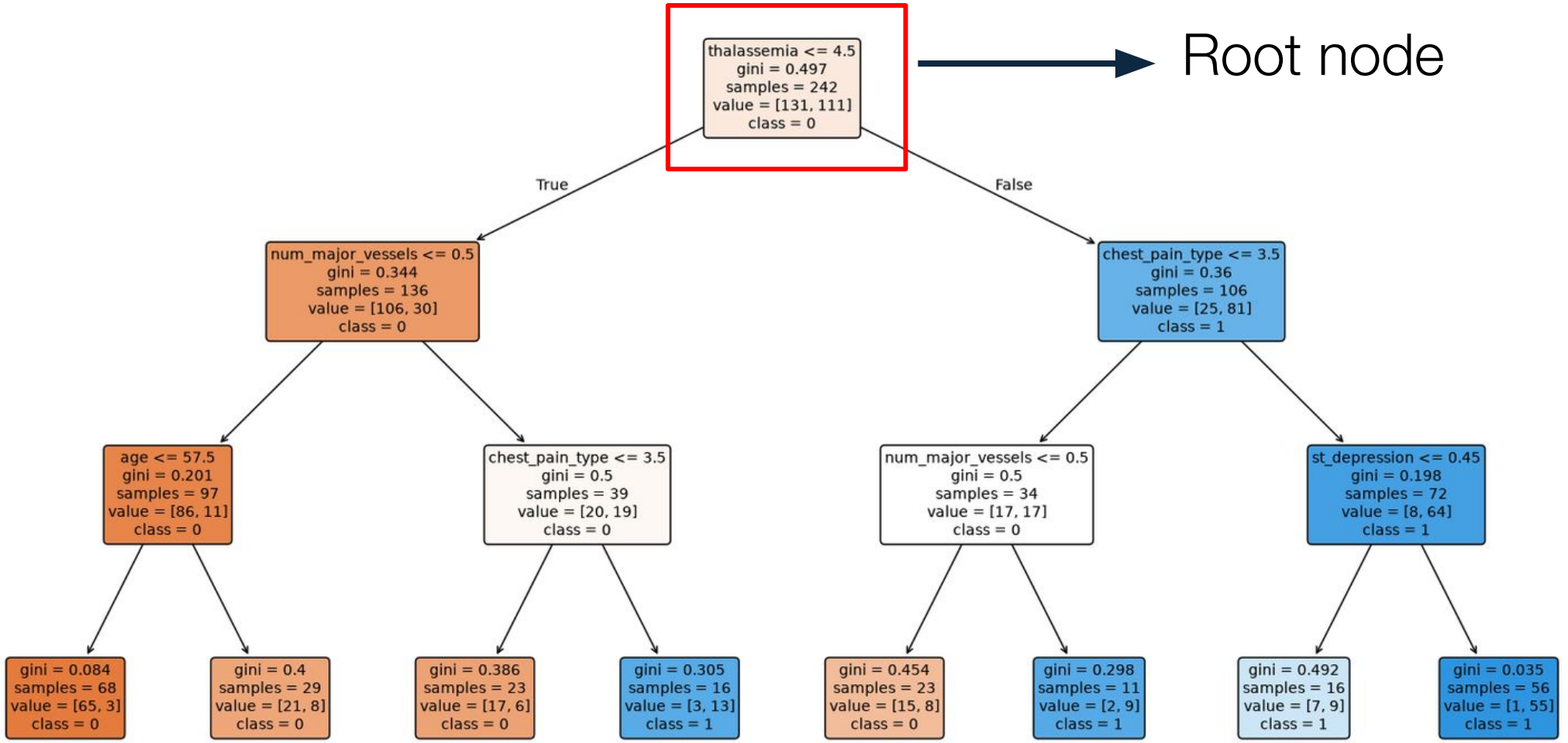
Module 2: Classification using tree-based algorithms

Gustavo Sganzerla Martinez

Machine Learning Foundations, Halifax, NS

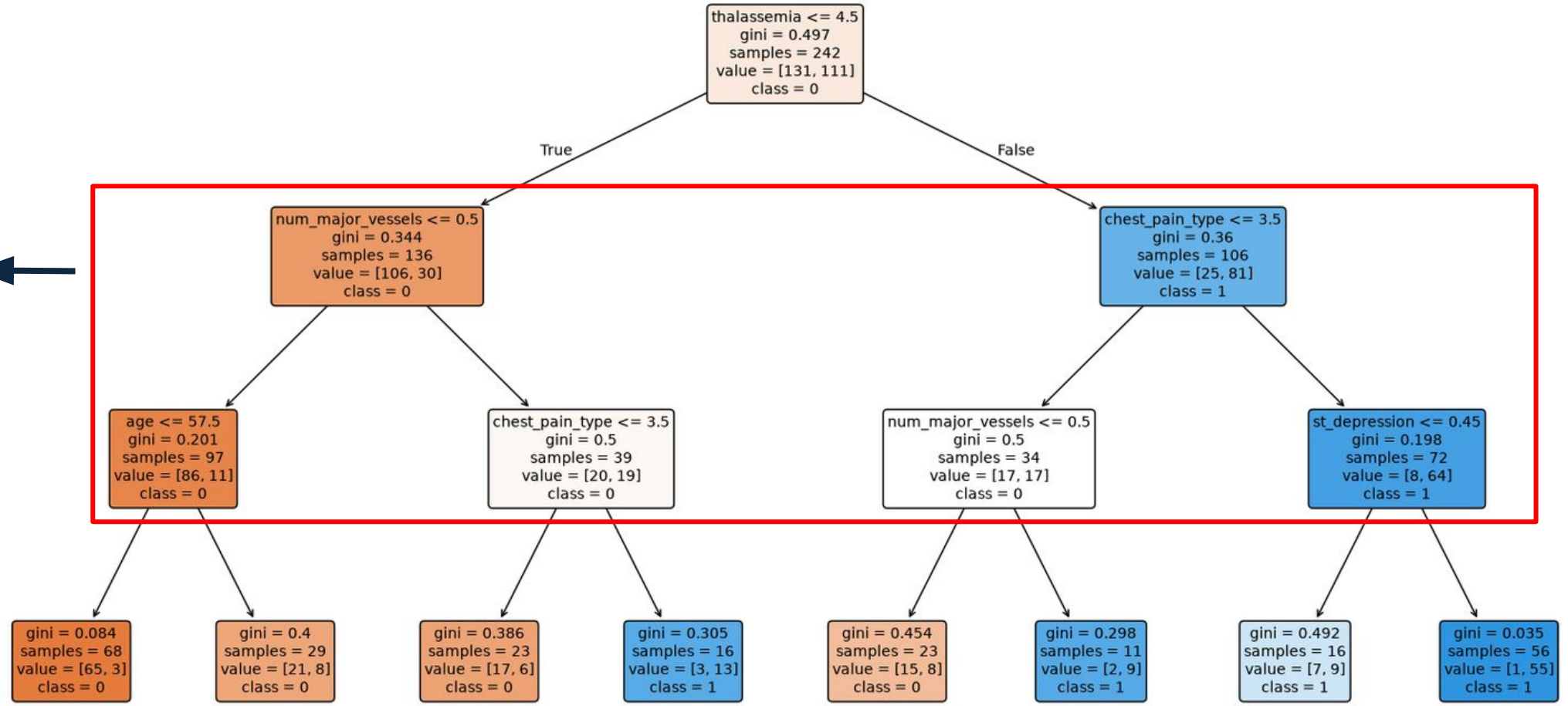
October 14-16, 2026

Structure of a Decision Tree

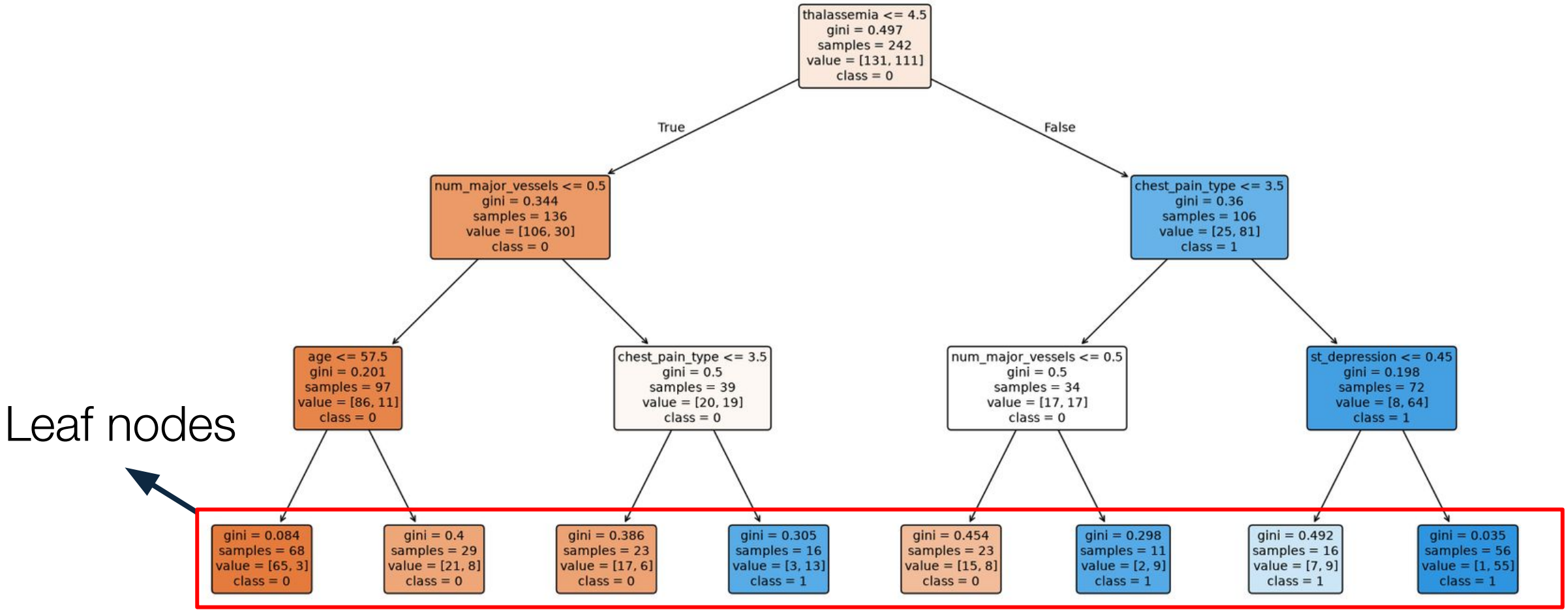


Structure of a Decision Tree

Internal nodes



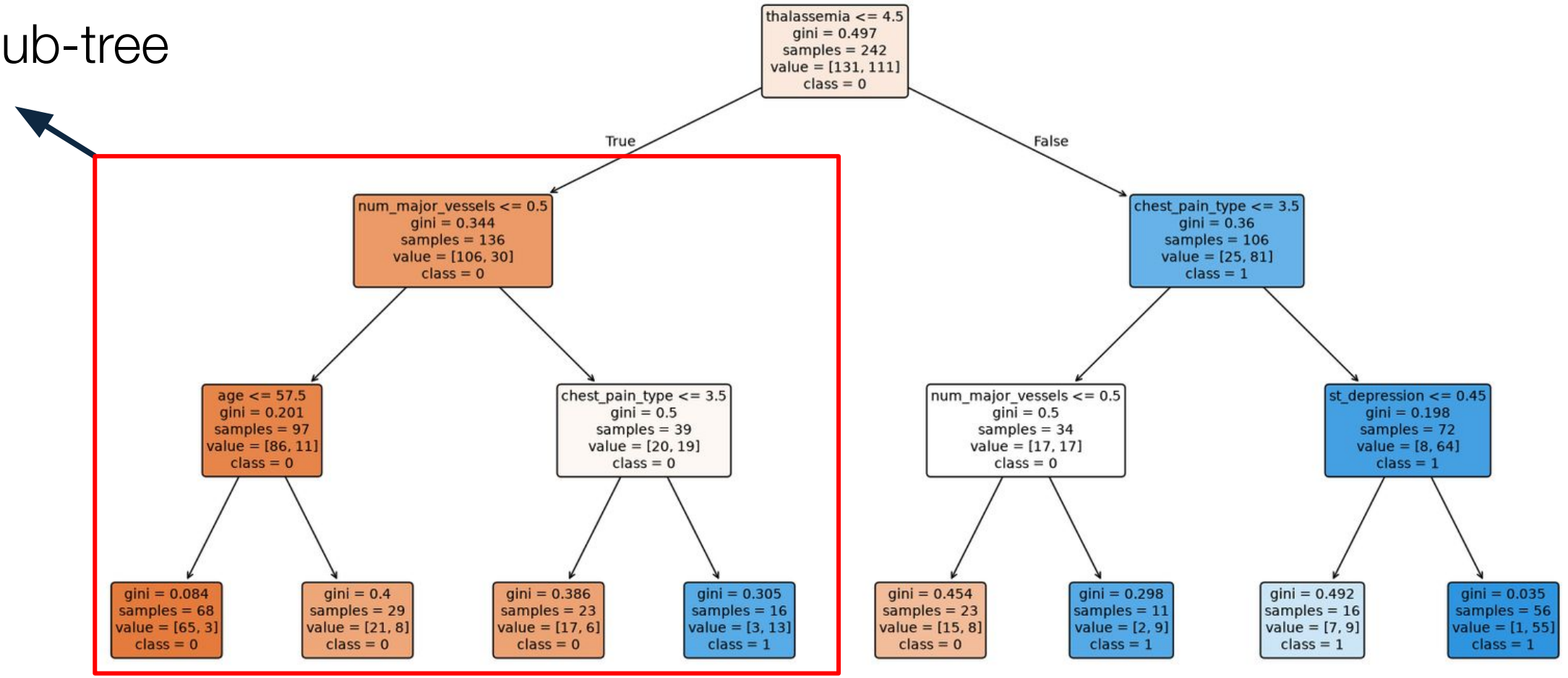
Structure of a Decision Tree



Leaf nodes

Structure of a Decision Tree

Sub-tree

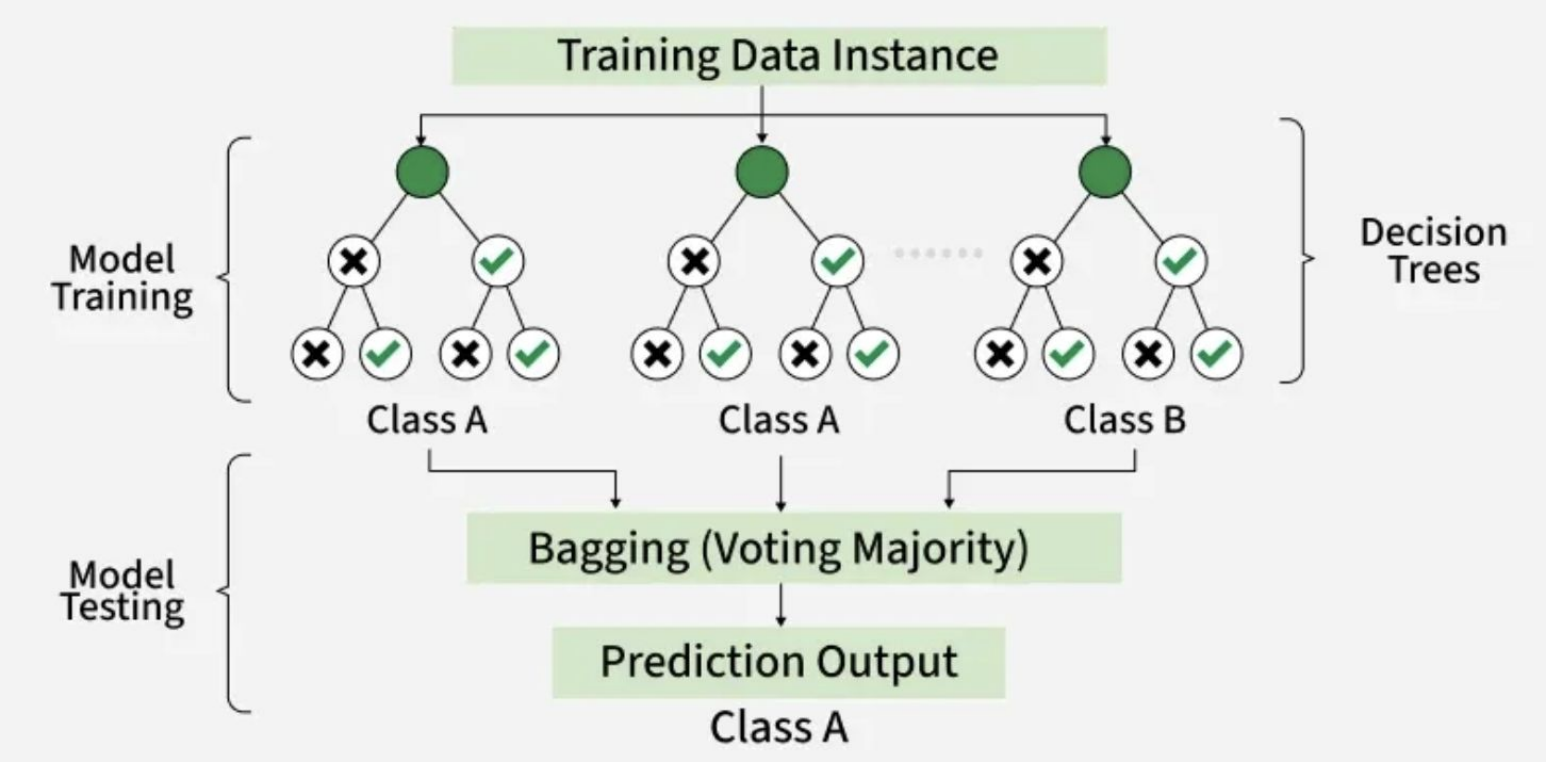


Key parameters of a Decision Tree

- **criterion**: function to measure the quality of a split (gini [default]; log_loss, entropy).
- **max_depth** = maximum depth of a tree (how many decision nodes vertically).
- **min_samples_split** = minimum number of samples required to split an internal node.
- **min_samples_leaf** = minimum number of samples required to be at a leaf node.

Random Forest

- Ensemble of multiple decision trees.
- Each tree is built with different subsets of X.



Key parameters of a Random Forest

- **n_estimators =** numbers of trees in the forest (default = 100).
- **max_features =** (sqrt, log2) The number of features to consider in each split.
- **criterion, max_depth, min_samples_split, min_samples_leaf** = Same as in Decision Trees.

Coffee Break & Networking Session

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